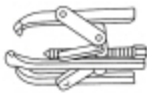


Engine

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)	
 ST1184-A	3 Jaw Puller 303-D121
 ST2132-A	Adapter Set, Step Plate 205-DS011 or equivalent
 ST1337-A	Installer, Connecting Rod 303-442 (T93P-6136-A)
 ST1377-A	Lifting Bracket, Engine 303-F047 (014-00073) or equivalent
 ST1385-A	Remover, Oil Seal 303-409 (T92C-6700-CH)
 ST1382-A	Remover, Crankshaft Rear Oil Seal 303-519 (T95P-6701-DH)
 ST1185-A	Slide Hammer 100-001 (T50T-100-A)

	Support Brackets, Engine 303-1507 or equivalent
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Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® Silicone Gasket Remover ZC-30-A	—

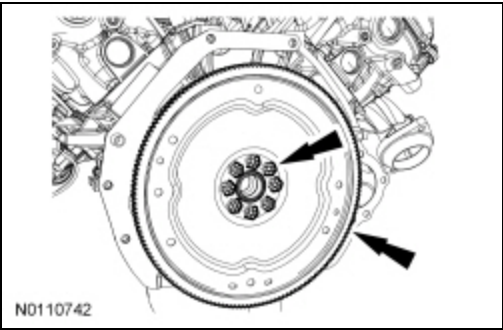
NOTICE: Remove the cylinder heads before removing the crankshaft. Failure to do so may result in engine damage.

NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

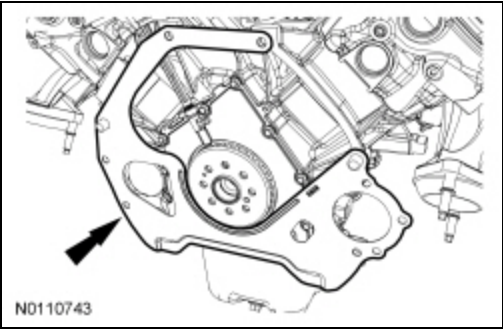
NOTE: The flexplate and crankshaft rear seal must be removed before mounting the engine on the engine stand.

NOTE: For additional information, refer to the exploded view under the Assembly procedure in this section.

1. Remove the 8 bolts and the flexplate.
 - Discard the 8 flexplate bolts.

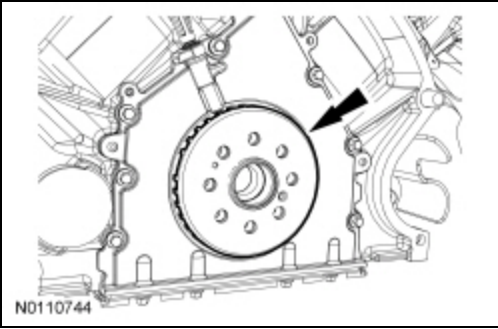


2. Remove the spacer plate.

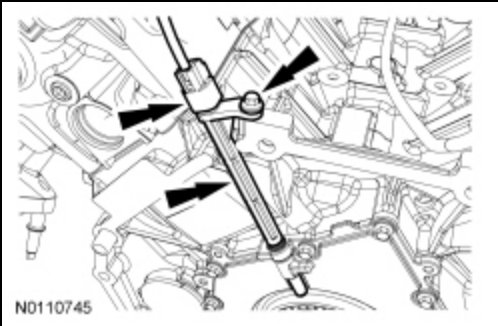


3. **NOTE:** Inspect the ignition pulse ring for damage. If the ignition pulse ring has been dropped or has any visual damage, it must be discarded.

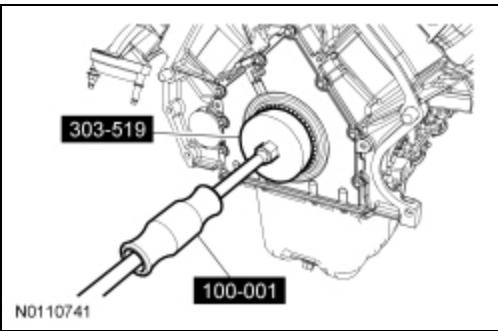
Remove the crankshaft ignition pulse ring.



4. Disconnect the electrical connector and remove the bolt and the Crankshaft Position (CKP) sensor.

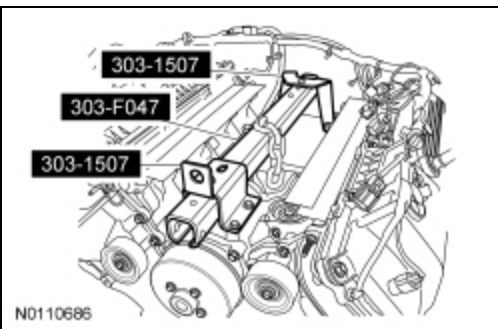


5. Using the Slide Hammer and the Crankshaft Rear Oil Seal Remover, remove and discard the crankshaft rear seal.

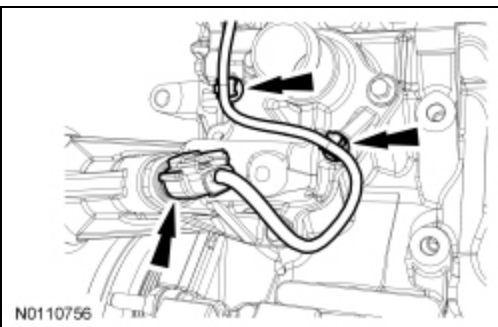


6. Mount the engine on a work stand.

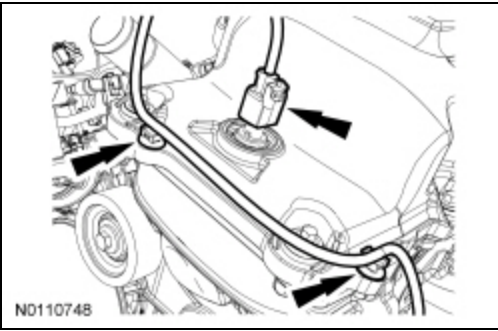
7. Remove the Engine Lifting Bracket and Engine Support Brackets.



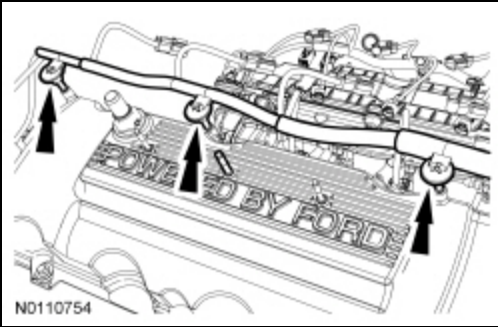
8. Disconnect the Engine Oil Pressure (EOP) switch electrical connector and wiring harness retainers.



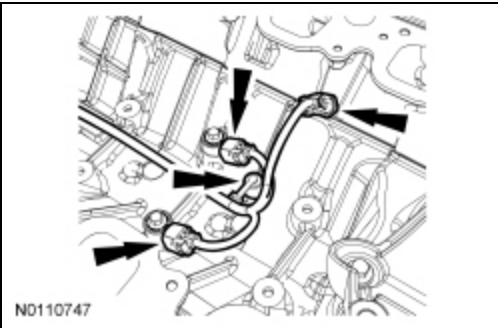
9. Disconnect the LH Variable Camshaft Timing (VCT) variable force solenoid electrical connector and the 2 wiring harness retainers.



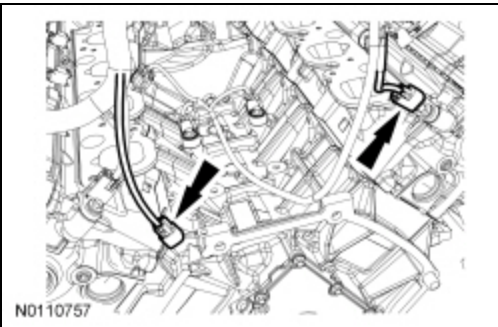
10. Disconnect the 3 engine wiring harness position retainers from the LH valve cover.



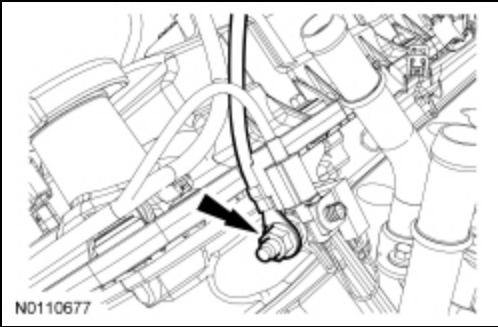
11. Disconnect the wiring harness retainer, the Knock Sensor (KS) and Cylinder Head Temperature (CHT) sensor electrical connectors.



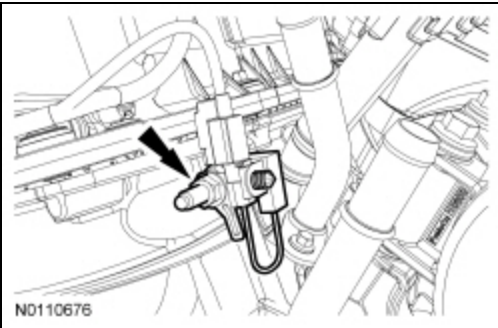
12. Disconnect the LH and RH Camshaft Position (CMP) electrical connectors.



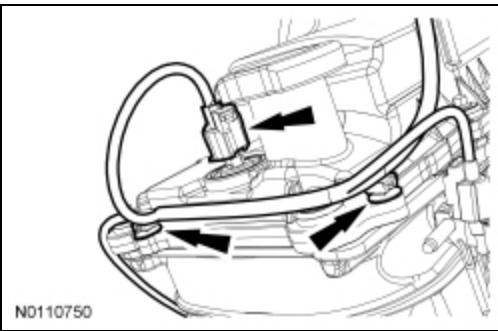
13. Remove the nut and the wiring harness ground cable.



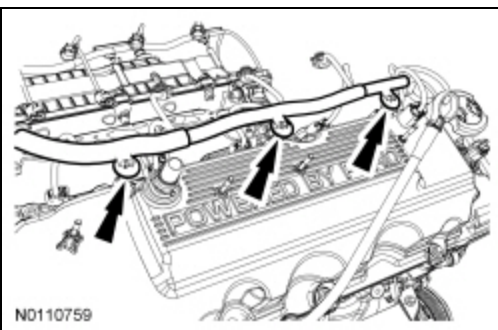
14. Remove the nut and the radio interference capacitor.



15. Disconnect the RH ~~YCT~~ variable force solenoid electrical connector and the 2 wiring harness retainers.

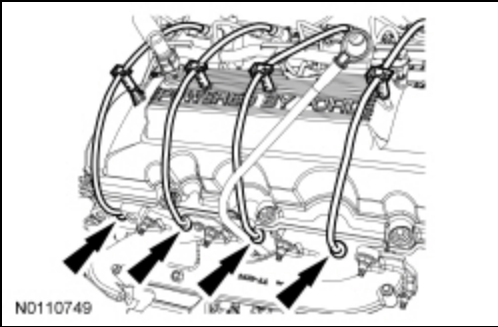


16. Disconnect the 3 engine wiring harness position retainers from the RH valve cover and remove the engine wiring harness.

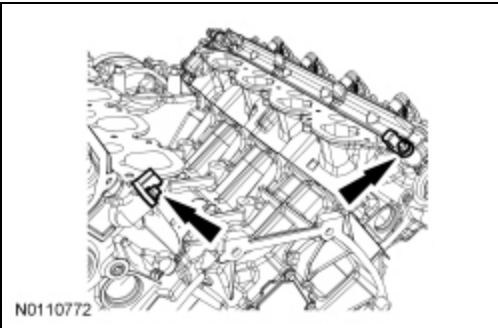


17. **NOTE:** *RH shown, LH similar.*

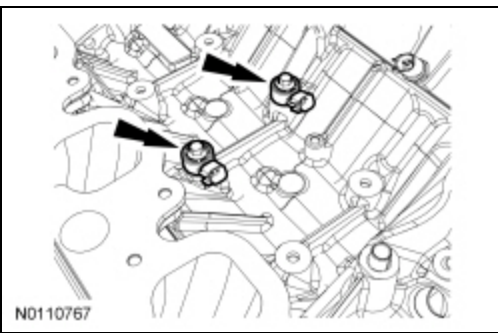
Disconnect the 8 ignition wires from the valve cover retainers and the lower spark plugs.



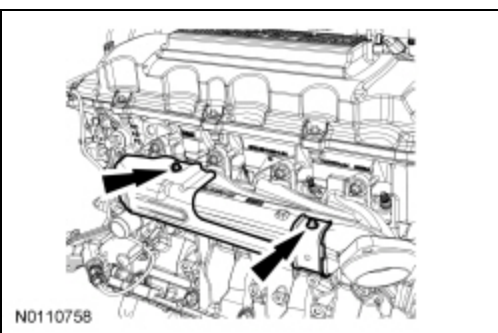
18. Remove the 2 bolts and the LH and RH CMP sensors.



19. Remove the 2 bolts and the RH and LH KS .

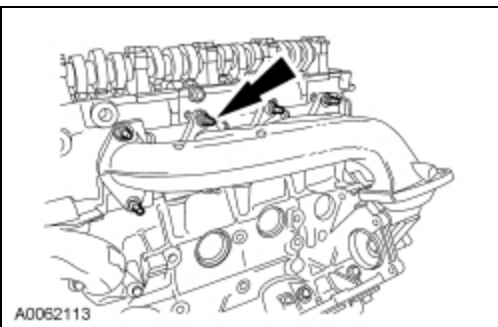


20. Remove the 2 bolts and the LH exhaust manifold heat shield.

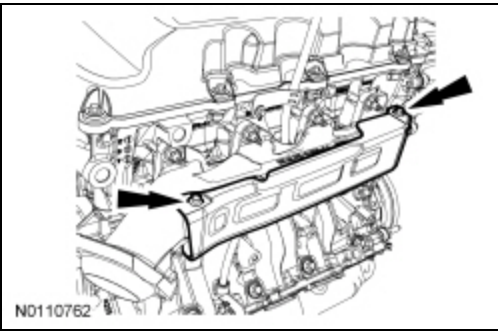


21. Remove the 8 nuts, the 8 studs, the LH exhaust manifold and the 2 gaskets.

- Discard the gaskets, nuts and studs.
- Inspect the exhaust manifold. For additional information, refer to Section 303-00.

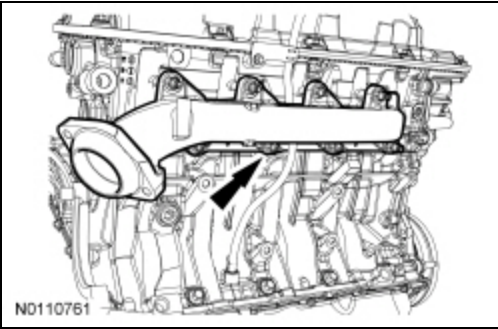


22. Remove the 2 bolts and the RH exhaust manifold heat shield.

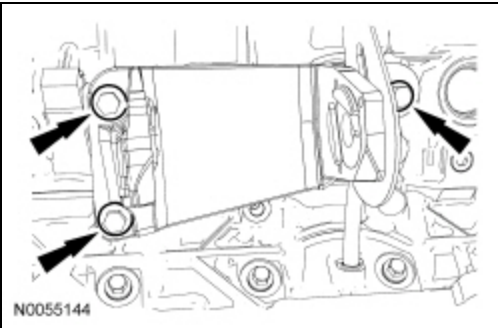


23. Remove the 8 nuts, the 8 bolts, the RH exhaust manifold and the 2 gaskets.

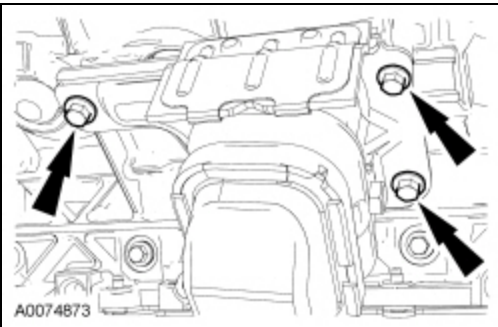
- Discard the gaskets, nuts and studs.
- Inspect the exhaust manifold. For additional information, refer to Section 303-00.



24. Remove the 3 bolts and the LH engine support insulator-to-cylinder block bracket.

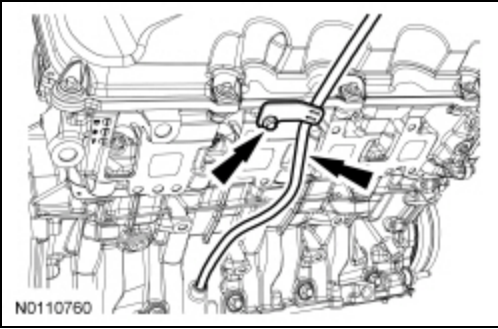


25. Remove the 3 bolts and the RH engine support insulator.



26. Remove the bolt and the oil level indicator tube.

- Discard the O-ring seal.



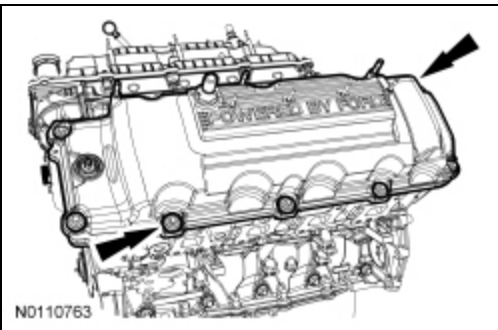
27. NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

NOTICE: Remove the valve cover carefully, or the Variable Camshaft Timing (VCT) variable force solenoid may be damaged.

NOTE: The bolts are part of the valve cover and should not be removed.

Loosen the 10 bolts and remove the LH valve cover.

- Clean the valve cover mating surface of the cylinder head with silicone gasket remover and metal surface prep. Follow the directions on the packaging.
- Discard the valve cover gasket. Clean the valve cover gasket groove with soap and water or a suitable solvent.



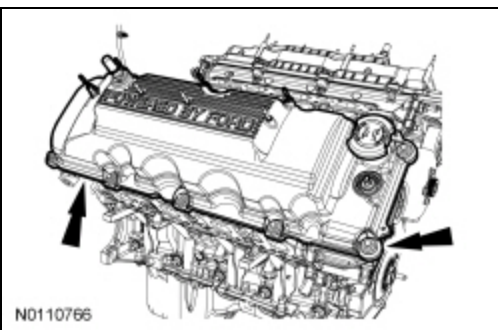
28. NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

NOTICE: Remove the valve cover carefully, or the Variable Camshaft Timing (VCT) variable force solenoid may be damaged.

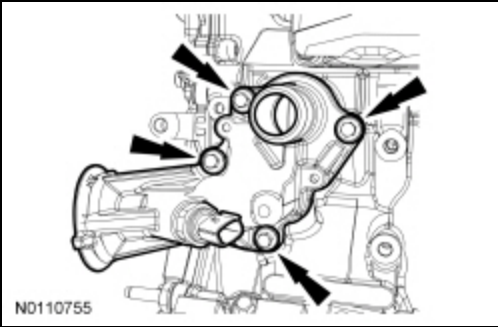
NOTE: The bolts are part of the valve cover and should not be removed.

Loosen the 10 bolts and remove the RH valve cover.

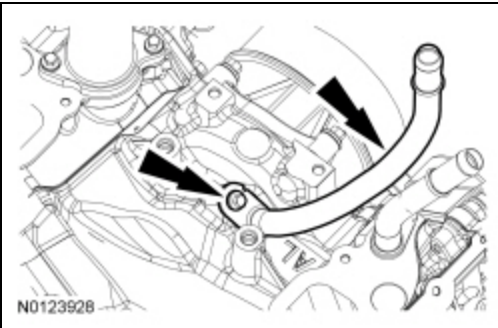
- Clean the valve cover mating surface of the cylinder head with silicone gasket remover and metal surface prep. Follow the directions on the packaging.
- Discard the valve cover gasket. Clean the valve cover gasket groove with soap and water or a suitable solvent.



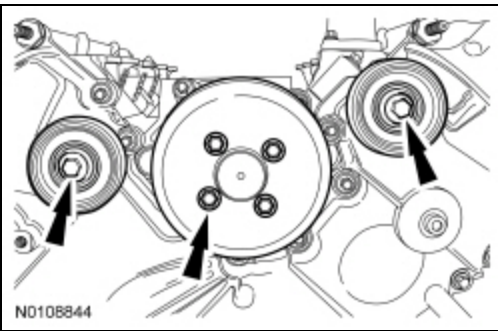
29. Remove the 4 bolts and the oil filter adapter.



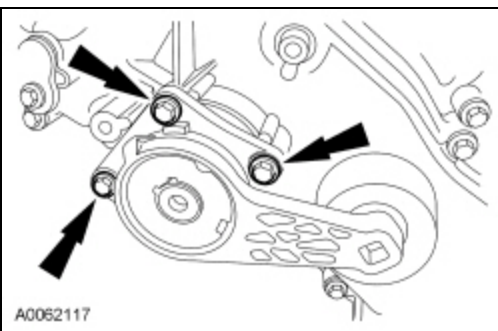
30. Remove the bolt and the heater coolant tube.



31. Remove the 6 bolts, coolant pump pulley and the 2 accessory drive belt idler pulleys.

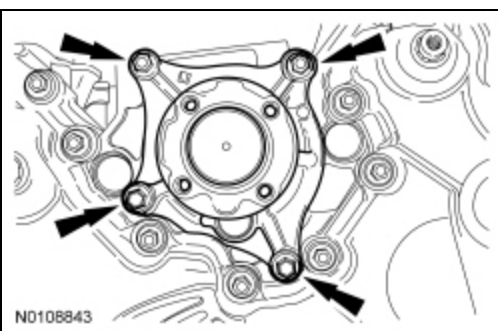


32. Remove the 3 bolts and the accessory drive belt tensioner.



33. Remove the 4 bolts and the coolant pump from the cylinder block.

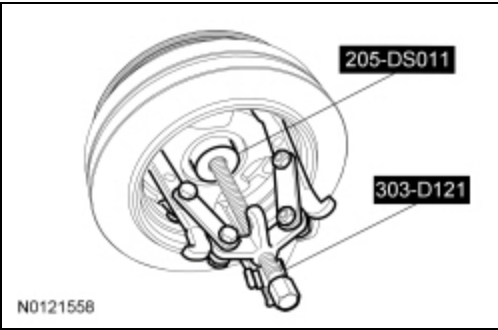
- Discard the O-ring seal.



34. Remove the crankshaft pulley bolt.

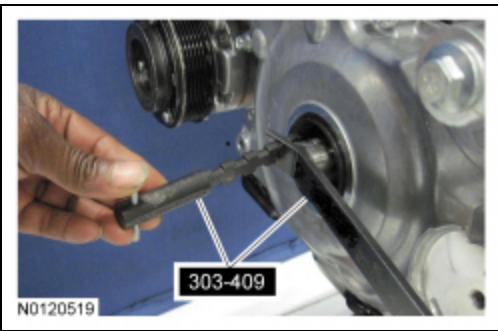
- Discard the crankshaft pulley bolt.

35. Using the 3 Jaw Puller and Step Plate, remove the crankshaft pulley.



36. **NOTICE:** Use care not to damage the engine front cover or the crankshaft when removing the seal.

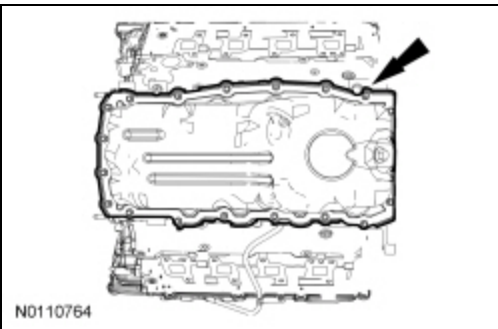
Using the Oil Seal Remover, remove the crankshaft front oil seal.



37. Remove the 20 bolts, oil pan and oil pan gasket.

- **NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges, which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

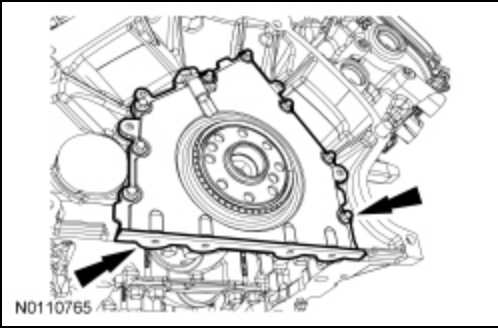
Clean and inspect the sealing surfaces.



38. **NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges, which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

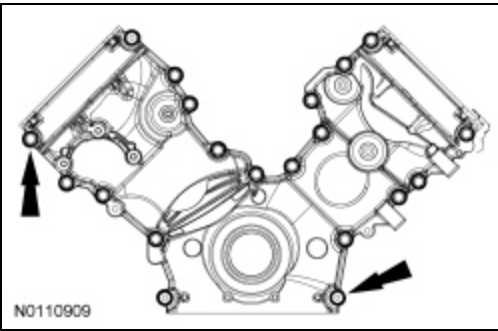
Remove the 8 bolts and the crankshaft rear seal retainer plate.

- Clean and inspect the sealing surfaces.



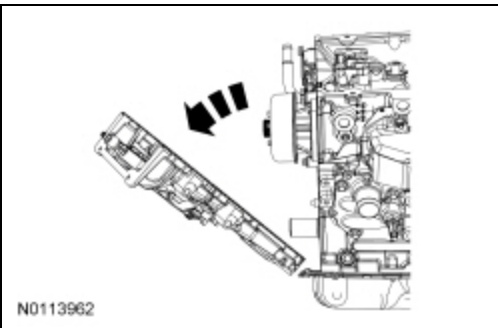
39. **NOTE:** *Correct fastener location is essential for assembly procedure. Record fastener location.*

Remove the 17 bolts and the 3 studs.



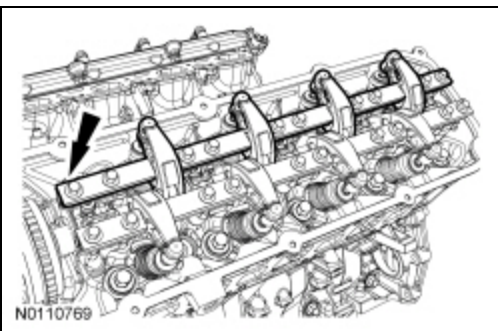
40. Remove the engine front cover from the front cover-to-cylinder block dowel.

- Remove and discard the engine front cover gaskets.



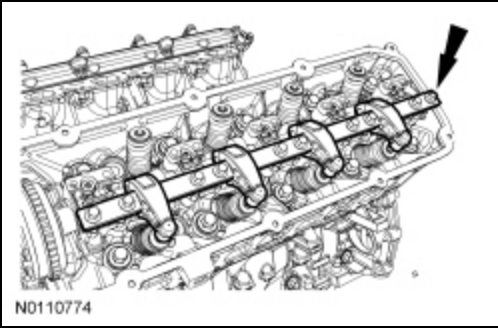
41. Rotate the engine clockwise until the No. 1 intake valve is open.

42. Remove the 10 bolts and the LH intake rocker arm shaft assembly.



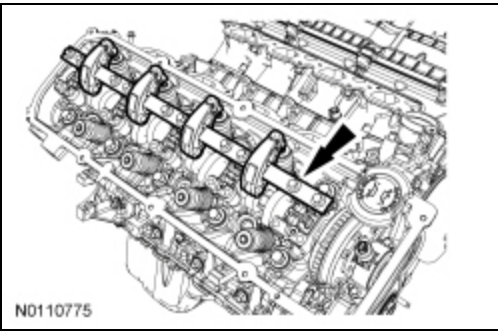
43. Rotate the engine clockwise until the No. 1 exhaust valve is open.

44. Remove the 10 bolts and the LH exhaust rocker arm shaft assembly.



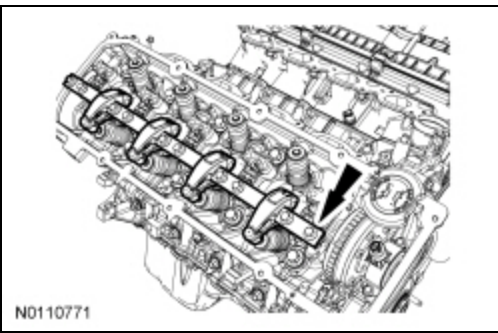
45. Rotate the engine clockwise until the No. 1 intake valve is closed.

46. Remove the 10 bolts and the RH intake rocker arm shaft assembly.

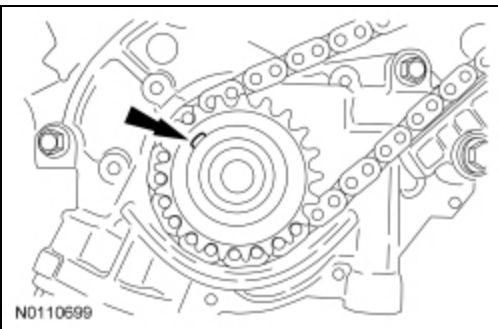


47. Rotate the engine clockwise until the No. 1 exhaust valve is closed.

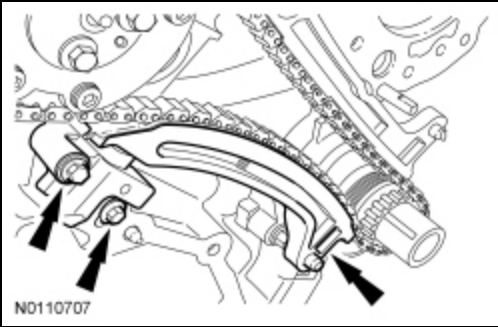
48. Remove the 10 bolts and the RH exhaust rocker arm shaft assembly.



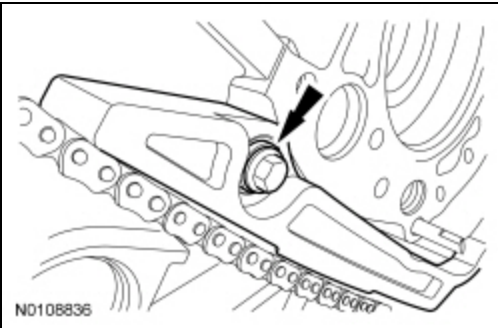
49. Position the crankshaft keyway at the 11 o'clock position.



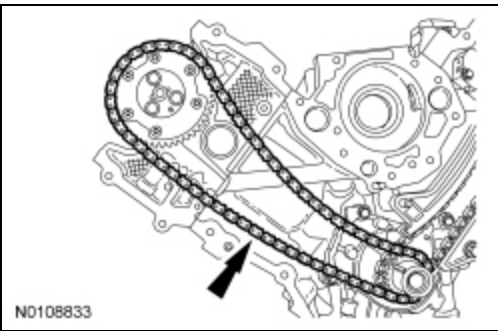
50. Remove the 2 bolts, the RH timing chain tensioner and tensioner arm.



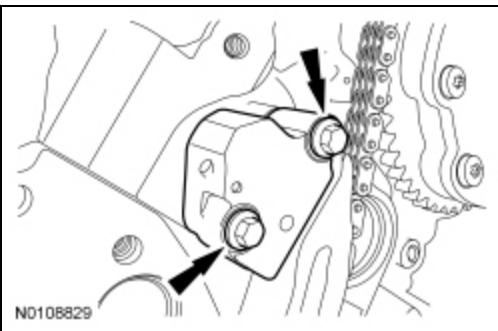
51. Remove the bolt and the RH timing chain guide.



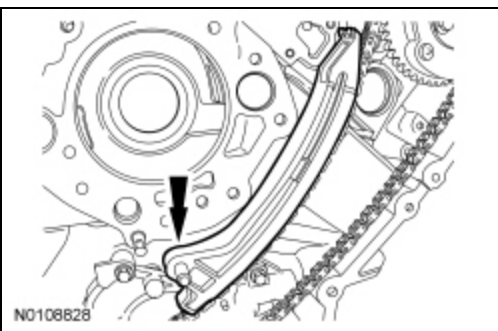
52. Remove the RH timing chain.



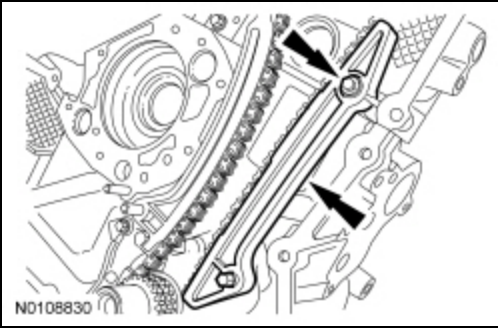
53. Remove the 2 bolts and the LH timing chain tensioner.



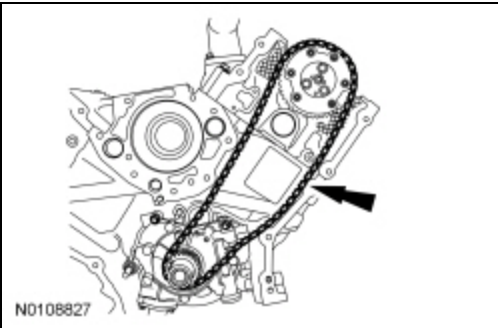
54. Remove the LH timing chain tensioner arm.



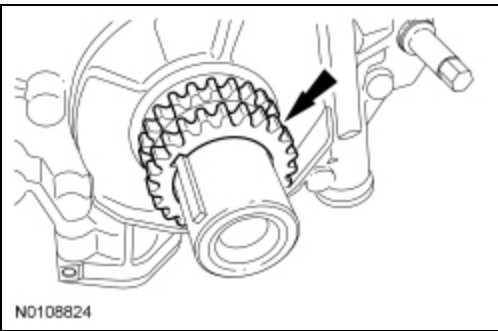
55. Remove the bolt and the LH timing chain guide.



56. Remove the LH timing chain.



57. Remove the crankshaft sprocket.



58. **NOTICE:** The cylinder head must be cool before removing it from the engine. Cylinder head warpage may result if a warm or hot cylinder head is removed.

NOTICE: Place clean shop towels over exposed engine cavities. Carefully remove the towels so foreign material is not dropped into the engine. Failure to follow this instruction may cause engine damage.

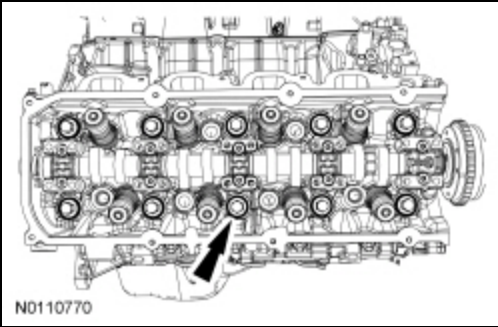
NOTICE: The cylinder head bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.

NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges, which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

NOTICE: Aluminum surfaces are soft and can be scratched easily. Never place the cylinder head gasket surface, unprotected, on a bench surface. Failure to follow this instruction may cause engine damage.

Remove the 10 bolts and the RH cylinder head.

- Discard the cylinder head gasket.
- Discard the cylinder head bolts.



59. NOTICE: The cylinder head must be cool before removing it from the engine. Cylinder head warpage may result if a warm or hot cylinder head is removed.

NOTICE: Place clean shop towels over exposed engine cavities. Carefully remove the towels so foreign material is not dropped into the engine. Failure to follow this instruction may cause engine damage.

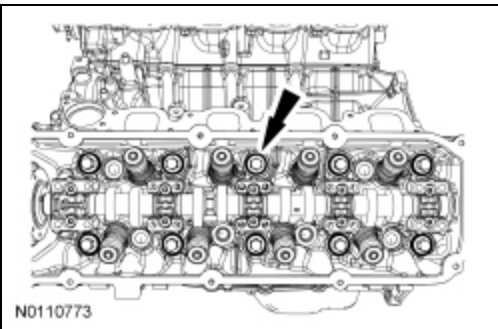
NOTICE: The cylinder head bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.

NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges, which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

NOTICE: Aluminum surfaces are soft and can be scratched easily. Never place the cylinder head gasket surface, unprotected, on a bench surface. Failure to follow this instruction may cause engine damage.

Remove the 10 bolts and the LH cylinder head.

- Discard the cylinder head gasket.
- Discard the cylinder head bolts.



60. NOTICE: Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges, which make leak paths. Use a plastic scraping tool to remove all traces of old sealant.

NOTICE: Observe all warnings or cautions and follow all application directions contained on the packaging of the silicone gasket remover and the metal surface prep. Failure to follow this procedure may cause engine damage.

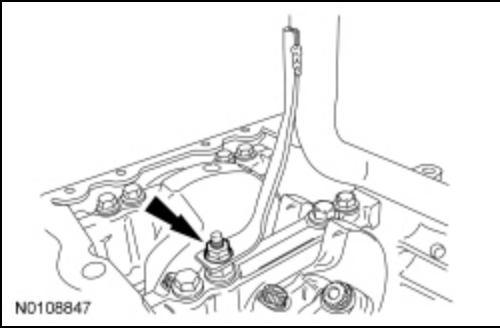
NOTE: *If there is no residual gasket material present, metal surface prep can be used to clean and prepare the surfaces.*

Clean the cylinder head-to-cylinder block mating surfaces of both the cylinder head and the cylinder block.

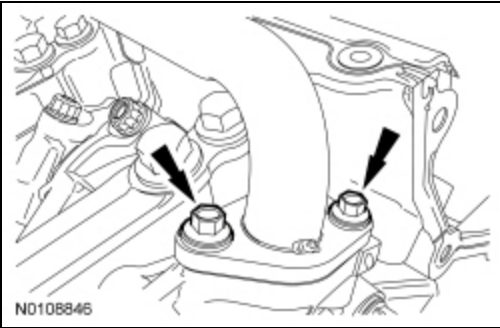
1. Remove any large deposits of silicone or gasket material with a plastic scraper.
2. Apply silicone gasket remover, following package directions, and allow to set for several minutes.
3. Remove the silicone gasket remover with a plastic scraper. A second application of silicone gasket remover may be required if residual traces of silicone or gasket material remain.
4. Apply metal surface prep, following package directions, to remove any remaining traces of oil or coolant, and to prepare the surfaces to bond with the new gasket. Do not attempt to make the metal shiny. Some staining of the metal surfaces is normal.

61. Support the cylinder heads on a bench with the head gasket side up. Check the cylinder head distortion and the cylinder block distortion, paying particular attention to the oil pressure feed area. For additional information, refer to Section 303-00.

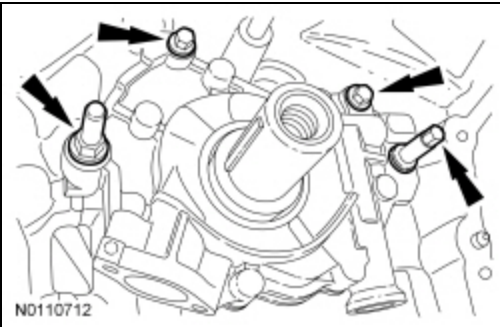
62. Remove the oil pump screen and pickup tube support bracket nut.



63. Remove the 2 bolts and the oil pump screen and pickup tube.



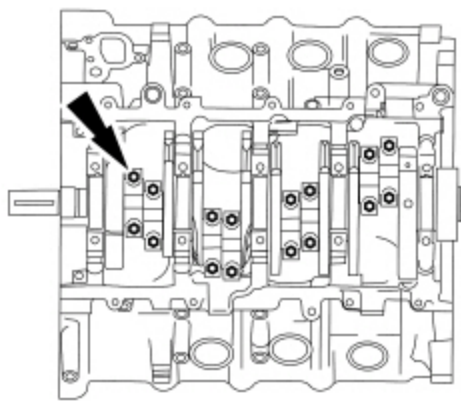
64. Remove the 2 upper bolts, the 2 lower stud bolts and the oil pump.



65. Before removing the pistons, inspect the top of the cylinder bores. If necessary, remove the ridge or carbon deposits from each cylinder using an abrasive pad or equivalent, following the manufacturer instructions.

66. **NOTICE: Verify that the connecting rods and rod caps have orientation numbers cast into them. If not, number the connecting rods and rod caps for correct orientation. Failure to follow these instructions may result in engine damage.**

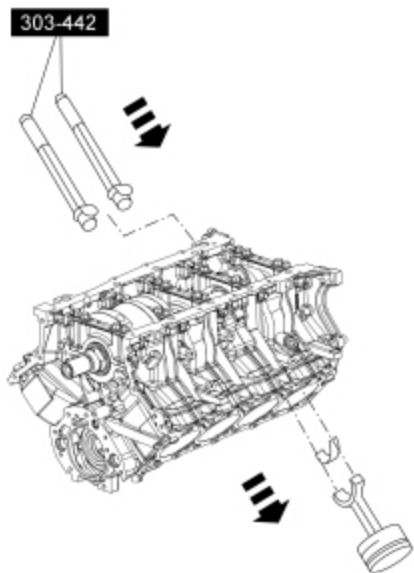
Remove the 16 bolts and the 8 connecting rod caps and bearings. Discard the bolts.



DA0578-A

67. **NOTICE:** Do not scratch the cylinder walls or crankshaft journals with the connecting rod. Failure to follow this instruction may result in engine damage.

Use the Connecting Rod Installer to push the piston through the top of the cylinder block.

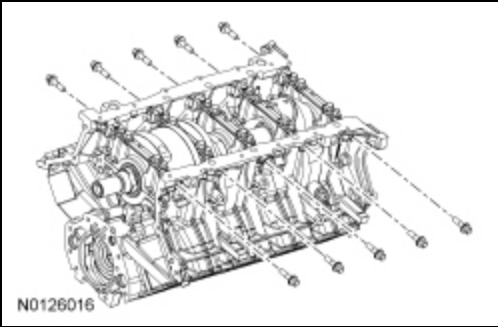


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68. Disassemble the 8 pistons. For additional information, refer to Piston in this section.

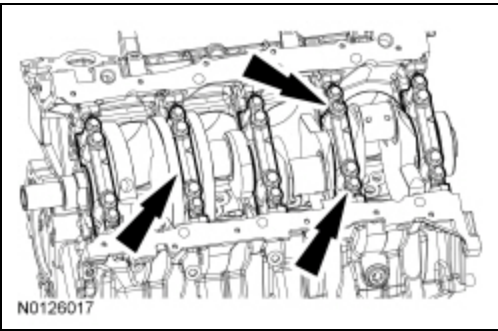
- Inspect the pistons. For additional information, refer to Section 303-00.

69. Remove and discard the 10 cross-mounted main cap bolts.

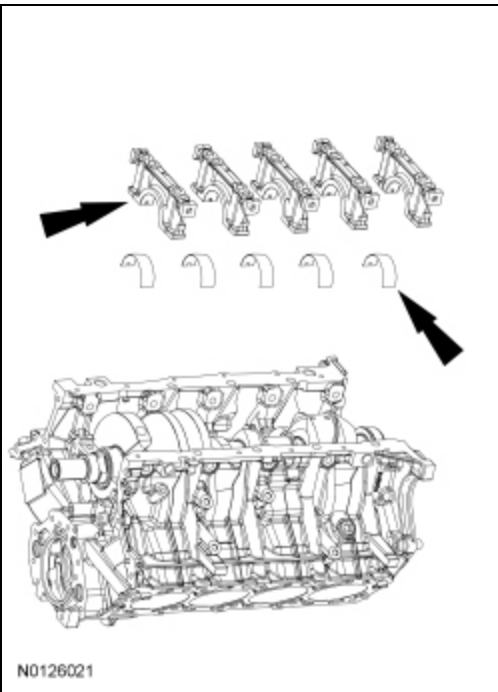


70. **NOTE:** *Correct stud bolt location is essential for assembly procedure. Record the stud bolt location.*

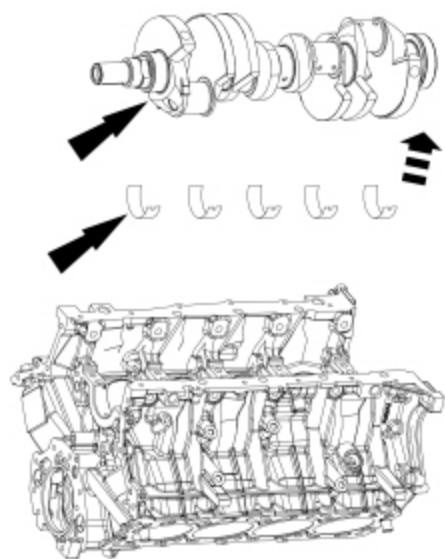
Remove and discard the 19 main bearing cap bolts and the stud bolt.



71. Remove the main bearing caps, the lower crankshaft main bearings and the lower thrust washer.

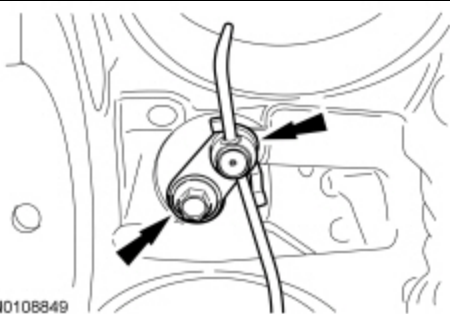


72. Remove the crankshaft, the upper crankshaft main bearings and the upper thrust washers from the cylinder block.



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73. Remove the 4 bolts and the 4 piston oil cooler valves.



N0108849

74. Inspect the cylinder block. For additional information, refer to Section 303-00.